HDR DATASET NOTE

Colon cancer adjuvant chemotherapy trial

HDR-US-0002 | version 1.0.0 | 2026-11-11

DOI	Pending, minted at publication
CREATORS	Imad El Badisy
COVERAGE	United States of America
PERIOD	1984 to 1987
UNIT	Patient event record
LICENCE	CC BY 4.0
LANDING PAGE	https://elbadisyimad.com/hdr/datasets/us-colon-cancer-adjuvant-trial/

Description

The Intergroup randomised trial led by the North Central Cancer Treatment Group and the Mayo Clinic: 929 patients with resected stage C colon cancer randomised to observation, levamisole, or levamisole plus fluorouracil, with tumour features and time to recurrence and to death.

A landmark adjuvant chemotherapy trial and a standard example for Cox models with recurrence and death as two outcomes per patient.

GOOD FOR

- Teaching the analysis of a three-arm randomised trial
- Models for recurrent or multiple events per patient

NOT GOOD FOR

- Evidence on current adjuvant treatment for colon cancer

Origin

ORGANISATION	North Central Cancer Treatment Group and Mayo Clinic
PUBLICATION	*Moertel CG et al. (1990). Levamisole and fluorouracil for adjuvant therapy of resected colon carcinoma. New England Journal of Medicine 322(6): 352-358*
COUNTRY	United States of America
ADDRESS	https://CRAN.R-project.org/package=survival
ACCESSED	2026-11-11
REUSE TERMS	Distributed openly for research in the survival R package (LGPL).
HOW OBTAINED	Taken from the BioBench R package (dataset colon), which took them from survival::colon.

Files

File	Content	Rows	Cols	Size
`colon.csv`	Two rows per patient, one for recurrence and one for death	1,858	16	80 kB
`us-colon-cancer-adjuvant-trial.xlsx`	All tables in one workbook, with About and Variables sheets			109 kB

SHA-256 checksums of every file are listed in `manifest.csv` on the landing page.

Variables

`colon.csv` Two rows per patient, one for recurrence and one for death

Variable	Type	Description
`id`	integer	Patient identifier
`study`	integer	Study (1 for all patients)
`rx`	string	Treatment arm (Obs observation, Lev levamisole, Lev+5FU levamisole plus fluorouracil)
`sex`	integer	Sex (1 male, 0 female)
`age`	integer	Age (years)
`obstruct`	integer	Obstruction of the colon by tumour (1 yes, 0 no)
`perfor`	integer	Perforation of the colon (1 yes, 0 no)
`adhere`	integer	Adherence to nearby organs (1 yes, 0 no)
`nodes`	integer	Number of lymph nodes with detectable cancer
`status`	integer	Status for this event type (1 event, 0 censored)
`differ`	integer	Tumour differentiation (1 well, 2 moderate, 3 poor)
`extent`	integer	Extent of local spread (1 submucosa, 2 muscle, 3 serosa, 4 contiguous structures)
`surg`	integer	Time from surgery to registration (0 short, 1 long)
`node4`	integer	More than 4 positive lymph nodes (1 yes, 0 no)
`time`	integer	Days from registration to the event or censoring
`etype`	integer	Event type (1 recurrence, 2 death)

Known issues

Read these before analysis.

1. **Demonstration entry.** Taken from the BioBench R package to show how HDR presents a dataset. It has not gone through HDR review and will be removed before launch.
2. **Two rows per patient,** one for recurrence (etype 1) and one for death (etype 2). Count patients with id, not rows.
3. **nodes and node4 disagree for some patients.** The survival package keeps both as received because the correct value is unknown.
4. **Missing values** in nodes (36 rows) and differ (46 rows).

How to cite

Imad El Badisy (2026). Colon cancer adjuvant chemotherapy trial (version 1.0.0) [Data set]. HDR: Health Data for Research. https://elbadisyimad.com/hdr/datasets/us-colon-cancer-adjuvant-trial/

Please also cite the origin: North Central Cancer Treatment Group and Mayo Clinic, Moertel CG et al. (1990). Levamisole and fluorouracil for adjuvant therapy of resected colon carcinoma. New England Journal of Medicine 322(6): 352-358.

Versions

Version	Date	Changes
1.0.0	2026-11-11	Demonstration release

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